

Introduction to Machine Learning (Study Guide)

Topic 1: Supervised Learning

Supervised learning maps inputs to outputs from labeled training examples. The algorithm analyzes labeled data and produces an inferred function for mapping new examples.

Topic 2: Unsupervised Learning

Unsupervised learning finds patterns in unlabeled data, building a compact internal representation without guidance.

Topic 3: Overfitting and Underfitting

Overfitting occurs when a model fits training data too well, capturing noise and failing to generalize. Underfitting occurs when a model is too simple to capture the trend. Study the bias-variance trade-off, which underlies both.